

GENAI STUDY ASSISTANT

Multi-Step Generative AI Workflow for Intelligent Learning — Without RAG

STUDENT	MULINTI SUJITH REDDY
REGISTER NO.	99230041076
DEPARTMENT	CSE(AIML)
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PROJECT	GenAI Study Assistant

01 | Concept

Purpose: The GenAI Study Assistant helps students understand academic topics, organize study material, generate explanations, and prepare for examinations. This version uses a multi-step prompt workflow without RAG, vector databases, embeddings, or automatic document retrieval.

Problem: A single large prompt may produce inconsistent study material. The proposed workflow separates question understanding, learning planning, content generation, review, and learning support so each stage has a clear role.

02 | Objective

The objective is to build a simple reusable GenAI workflow in which every stage depends on the previous stage. The assistant should generate student-friendly learning content, check its quality, and produce a refined final answer.

03 | Workflow Overview

1. Understand	2. Plan	3. Generate	4. Review	5. Learn
Question	Study approach	Explanation	Quality check	Quiz / Summary

04 | Workflow Steps

- **Question Understanding:** identify subject, topic, difficulty, learning objective, and expected answer format.
- **Learning Plan:** use the analysis to decide concepts, examples, key points, and response structure.
- **Content Generation:** use the plan to generate clear explanations and exam-oriented study material.
- **Review & Refinement:** evaluate the draft against explicit criteria and correct weaknesses.
- **Learning Support:** turn the refined content into summaries, flashcards, MCQs, or revision questions.

05 | Multi-Step Prompt Workflow

Demonstration question: “Explain deadlock prevention in simple terms and give me important points for an exam.”

Step 1 — Understand & Plan

Act as an academic study-planning assistant. Analyze the student's question. Identify the subject, topic, learning objective, difficulty level, important concepts, and preferred answer format. Do not create the final answer yet. Return: (1) learning objective, (2) key concepts, (3) response structure, and (4) requirements that should be passed to the next step.

Student question: “Explain deadlock prevention in simple terms and give me important points for an exam.”

OUTPUT — Step 1

Subject: Operating Systems

Topic: Deadlock Prevention

Learning objective: Understand what deadlock prevention means and remember the main prevention methods for an examination.

Difficulty: Beginner / undergraduate level

Key concepts: deadlock, prevention, resource-allocation conditions, methods for preventing deadlock

Response structure: simple definition → explanation → important prevention methods → exam key points → short recap.

Next-step requirement: explain the topic in simple language and emphasize points useful for revision.

Step 2 — Generate Study Content

Using the complete Step 1 output, create a clear and student-friendly study response for the original question. Follow the learning objective, key concepts, difficulty level, and response structure from Step 1. Include simple explanations and exam-oriented points. Avoid unnecessary complexity.

STEP 1 OUTPUT:

[The complete Step 1 output above]

OUTPUT — Step 2

Deadlock Prevention — Simple Explanation

Deadlock prevention is a method of designing a system so that at least one condition required for a deadlock cannot occur. A deadlock happens when processes are permanently waiting for resources held by one another.

Important idea: prevent the system from satisfying all the conditions that can lead to deadlock.

Exam points:

1. Deadlock occurs when processes wait indefinitely for resources.
2. Prevention works by restricting resource allocation so a required deadlock condition is broken.
3. The approach is proactive: the system tries to stop deadlock before it occurs.
4. The answer should focus on the prevention conditions and the methods discussed in the course material.

Quick recap: Deadlock prevention = design resource allocation so deadlock cannot form.

06 | Evaluation / Refinement Gate

Step 3 does not create an unrelated answer. It takes the complete Step 2 output, checks it against explicit constraints, identifies weaknesses, and then improves that same content.

Step 3 — Critique & Refine

Review the complete Step 2 response against these constraints:

1. It must answer the original question.
2. It must be understandable to a student.
3. Important concepts must be covered without unnecessary complexity.
4. The organization must be easy to revise for an examination.
5. The answer must clearly distinguish the definition, explanation, key points, and recap.

First identify weaknesses or missing requirements. Then produce the corrected final answer. Briefly state what you improved.

STEP 2 OUTPUT:

[The complete Step 2 output above]

OUTPUT — Step 3: Critique

- The response is clear and directly answers the question.
- The explanation is suitable for a beginner, but the key prevention idea can be stated more directly.
- The exam points can be made more compact and revision-friendly.
- The final answer should end with a short memory aid.

Changes made: simplified the opening definition, strengthened the central prevention idea, organized the exam points, and added a concise memory aid.

OUTPUT — Step 3: Refined Final Answer

Deadlock Prevention

Deadlock prevention means designing the operating system so that a deadlock cannot occur. A deadlock occurs when processes wait indefinitely because each process is waiting for a resource held by another process.

How prevention works: The system controls resource allocation so that at least one condition necessary for deadlock is prevented.

Important exam points:

- Deadlock prevention is a proactive approach.
- It prevents the system from allowing all required deadlock conditions to exist together.
- Resource allocation and process requirements are controlled to avoid permanent waiting.
- The main goal is to make deadlock impossible rather than detect it after it happens.

Memory aid: Prevention = “stop deadlock before it forms.”

07 | Why the Prompts Are Chained

Dependency	Evidence in the workflow
Step 1 → Step 2	Step 2 explicitly uses the learning objective, key concepts, difficulty, and response structure produced by Step 1.
Step 2 → Step 3	Step 3 explicitly reviews the complete Step 2 response and improves it against a checklist.
Critique Gate	Step 3 first identifies weaknesses and then produces a corrected final answer.

08 | Prompt Chaining Architecture

STAGE	INPUT / CONTEXT	OUTPUT
1	Student question	Study requirement + learning plan
2	Step 1 output + question	Draft study content
3	Step 2 output + checklist	Critique + refined final answer
4	Refined answer	Summary / MCQs / flashcards

09 | Expected Final Learning Output

The final result is a concise, student-friendly explanation with important exam points and a short memory aid. Because the answer passes through an explicit review stage, the final version is easier to revise than the initial draft.

10 | Key Benefits

- ✓ Simple architecture without RAG, vector databases, embeddings, or retrieval pipelines
- ✓ Clear dependency between prompts
- ✓ Reusable for different academic subjects
- ✓ Supports summaries, flashcards, MCQs, and revision questions
- ✓ Evaluation stage improves clarity and organization
- ✓ Demonstrates practical prompt engineering and multi-step GenAI design

11 | Limitations

- The assistant depends on the question and study context supplied by the student.
- Without RAG, it does not automatically search a private document collection.
- Source-specific grounding is not guaranteed unless source material is supplied.
- Output quality depends on the underlying language model and prompt quality.

12 | Suggested Technology Stack

LAYER	TECHNOLOGY
Language	Python
LLM	OpenAI / Gemini / other approved LLM
Prompt Management	Python prompt templates / Jinja2
Interface	Streamlit or simple web interface
Evaluation	Fixed academic question set + quality checklist

13 | Assessment Submission Checklist

■	Three distinct prompts are executed sequentially in the same chat session.
■	Prompt 2 uses the actual output of Prompt 1.
■	Prompt 3 evaluates and refines the actual output of Prompt 2.
■	Intermediate AI outputs are documented.

14 | Conclusion

The GenAI Study Assistant demonstrates a practical multi-step Generative AI workflow. The project progresses from understanding the student's question to planning, generation, critique, and refinement. Each prompt depends on the previous stage, while the evaluation gate improves the clarity, organization, and usefulness of the final learning response.

15 | Conclusion

The GenAI Study Assistant demonstrates a practical multi-step Generative AI workflow. The project progresses from understanding the student's question to planning, generation, critique, and refinement. The documented outputs show how each prompt depends on the previous stage and how the evaluation gate improves the final learning response.